PSE Exam Solutions and Comments - 2019

Question 1 – Solutions and Comments

(a) - Solution:

The merit order dispatch signifies the procedure of dispatching the available generators on the basis of their bids or marginal costs and in a way that minimizes the total cost of supplying the demand. Once this least-cost dispatch has been determined, the system marginal price signifies the marginal cost of serving an additional unit of demand at the least-cost. In contrast with other commodities, electrical energy cannot be economically stored in large quantities and thus it must be produced exactly when it is consumed by the inflexible demand. Moreover, the electrical demand is not the same every hour, day, month etc and the generation side is characterized by a production cost which varies with the amount of demand. The combination of these effects explains why the system marginal price varies with time.

(a) - Comment:

Many students answered this question correctly. However, when explaining why the system marginal price varies with time in the case of electricity, some students did not consider the fluctuation of the demand levels from off-peak to peak values. In addition, some students did not state that the least-cost dispatch of the available generators is used to determine electricity price.

(b) - Solution:

In the optimal solution of the problem, the marginal cost of each generator is equal to the system marginal cost at each hour λ_t yielding:

$$\frac{\partial C_{1,t}(P_{1,t})}{\partial P_{1,t}} = 1 + 0.13 * P_{1,t} = \lambda_t \rightarrow P_{1,t} = \frac{\lambda_t - 1}{0.13} \quad (1)$$

$$\frac{\partial C_{2,t}(P_{2,t})}{\partial P_2} = 15 + 2 * P_{2,t} = \lambda_t \rightarrow P_{2,t} = \frac{\lambda_t - 15}{2} \quad (2)$$

The satisfaction of the system demand implies that:

$$P_{1,t} + P_{2,t} = D_t \quad (3)$$

Substituting (1)-(2) in (3) gives for the system marginal cost:

$$\frac{\lambda_t - 1}{0.13} + \frac{\lambda_t - 15}{2} = D_t \rightarrow \lambda_1 = 32.371 \left[\frac{£}{MWh}\right],\ \lambda_2 = 119.038 \left[\frac{£}{MWh}\right] \text{ and}$$

$$\lambda_3 = 43.357 \left[\frac{£}{MWh}\right] \quad (4)$$

Hour 2 is characterised by the highest system marginal cost, since it exhibits the highest level of demand.

Substituting (4) in (1) and (2) yields for the optimal production of each generator:

$$P_{1,1} = 241.315\ [MW],\ P_{1,2} = 907.981\ [MW] \text{ and } P_{1,3} = 325.822\ [MW] \quad (5)$$

$P_{2,1} = 8.685\ [MW], P_{2,2} = 52.019\ [MW]$ and $P_{2,3} = 14.178\ [MW]$ (6)

Substituting (5)-(6) in the cost functions of the generators yields for the total cost of operating the system over the 3-hour period:"

$$C_{tot} = \sum_i \sum_t C_{i,t}(P_{i,t}) = 73244.225[£] \quad (7)$$

The total profit for each generator Ω_i is given by the difference between its respective revenue and cost:

$$\Omega_1 = \sum_t [\lambda_t P_{1,t} - C_{1,t}(P_{1,t})] = 62293.451\ [£] \quad (8)$$

$$\Omega_2 = \sum_t [\lambda_t P_{2,t} - C_{2,t}(P_{2,t})] = 1572.418\ [£] \quad (9)$$

(b) - Comment:

Many students answered this question correctly but some students round the numbers of system marginal cost λ_1, λ_2, and λ_3 in equation (4), i.e. $\lambda_1 = 32\ \left[\frac{£}{MWh}\right]$, $\lambda_2 = 119\ \left[\frac{£}{MWh}\right]$ and $\lambda_3 = 43\ \left[\frac{£}{MWh}\right]$, leading to less accurate calculation of total cost and total profit of each generator.

(c) - Solution:

At hour 1, given that $P_{2,1} < P_2^{min}$ the solution calculated in (a) is not feasible when considering capacity limits of the generators. The optimal solution for hour 1 can now be calculated by fixing the output of generator 2 to:

$$P'_{2,1} = P_2^{min} = 50[MW] \quad (10)$$

and satisfying the remaining demand:

$$D'_1 = D_1 - P'_{2,1} = 250 - 50 = 200\ [MW] \quad (11)$$

By generator 1. Following the same method as in **(a)**, we obtain:

$$\frac{\lambda'_1 - 1}{0.13} = 200 \rightarrow \lambda'_1 = 27.00\ [\frac{£}{MWh}] \quad (12)$$

The optimal production of generator 1 equals to the remaining demand:

$$P'_{1,1} = 200.00\ [MW] \quad (13)$$

Similarly, at hour 2, given that $P_{1,2} > P_1^{max}$ the solution calculated in **(a)** is not feasible when considering given output limits of the generators. The optimal solution for hour 2 can now be calculated by fixing the output of generator 1 to:

$$P'_{1,2} = P_1^{max} = 900\ [MW] \quad (14)$$

and satisfying the remaining demand:

$$D'_2 = D_2 - P'_{1,2} = 60\ [MW] \quad (15)$$

by the generator 2. Following the same method as in **(a)**, we obtain:

$\frac{\lambda_2'-15}{2.0} = 60 \rightarrow \lambda_2' = 135.00\ [\frac{£}{MWh}]$ (16)

The optimal production of generator 2 equals the remaining demand:

$P_{2,2}' = 60\ [MW]$ (17)

Similarly, at hour 3, given that $P_{2,3} < P_2^{min}$ the solution calculated in **(a)** is not feasible when considering the given output limits of the generators. The optimal solution for hour 3 can now be calculated by fixing the output of generator 2 to:

$P_{2,3}' = P_2^{min} = 50\ [MW]$ (18)

and satisfying the remaining demand:

$D_3' = D_3 - P_{2,3}' = 290\ [MW]$ (19)

through the remaining generators 2. Following the same method as in **(a)**, we get:

$\frac{\lambda_3'-1}{0.13} = 290 \rightarrow \lambda_3' = 38.70\ [\frac{£}{MWh}]$ (20)

The optimal production of generator 1 equals the remaining demand:

$P_{1,3}' = 290\ [MW]$ (21)

Substituting (10), (12)-(14), (16)-(18), and (20)-(21) in the cost functions of the generators yields for the total cost of operating the system:

$C_{tot}' = \sum_i \sum_t C_{i,t}(P_{i,t}') = 76496.500\ [£]$ (22)

The profit for each generator is given by the difference between its respective revenue and cost:

$\Omega_1' = \sum_t [\lambda_t' P_{1,t}' - C_{1,t}(P_{1,t}')] = 74036.500\ [£]$ (23)

$\Omega_2' = \sum_t [\lambda_t' P_{2,t}' - C_{2,t}(P_{2,t}')] = -1025.00\ [£]$ (24)

(c) - Comment:

Some students could not find the marginal price in the system after considering the output limits. For the first case of $D_1 = 250\ [MW]$*, production of generation 2 should be changed to* $P_{2,1}' = P_2^{min} = 50[MW]$*, and the marginal price should be determined by generator 1* $\frac{\lambda_1'-1}{0.13} = 200 \rightarrow \lambda_1' = 27.00\ [\frac{£}{MWh}]$*. But some students kept the marginal price as* ***(b)*** $\lambda_1 = 32.371\ \left[\frac{£}{MWh}\right]$*. This led to incorrect calculation of the system total cost and the profit of each generator.*

(d) - Solution:

The negative profit of generator 2 means that its operating cost is larger than its revenue in the optimal solution and thus it experiences economic losses at the optimal solution (does not recover its operating costs). This is happening due to the combination of the high variable costs of generator 2 and its high minimum output limit with respect to the initial unconstrained dispatch calculated in **(b)**. Since the generators are free to participate in the market based on their profitability, generator 2
can avoid the negative profits (economic losses) by decommitting from the market.

(d) - Comment:

*Many students demonstrated the negative profit of generator 2 in quantitively analysis (i.e. high variable cost and low marginal cost), but did not explain the reason for the negative profit from the physical perspective (as generators are free to participate in the market based on their profitability, as assessed in **(b)**.*

Question 2 – Solutions and Comments

(a) - Solution:

A brief discussion of the following points is expected:

1) There tend to be fairly large participants able to influence the energy price by their actions.
2) The low elasticity of demand facilitates the abuse of market power.
3) Locational effects could prevent the free flow of the commodity.
4) Electricity networks are natural monopolies.

(a) - Comment:

Most of students could only provide two or three points in the above discussion.

(b) - Solution:

In a perfectly competitive market, no participant has the ability to influence the market price through its individual actions (i.e. all participants act as price takers). This assumption is valid if the number of market participants is large and if none of them control a large proportion of the production or consumption. Perfect competition is a highly desirable goal because it encourages efficient market behaviour. Strategic market players can manipulate the prices either by withholding quantity (physical withholding) or by raising the asking price (economic withholding), compared to the perfectly competitive case. In the presence of imperfect competition, market prices tend to be higher than producers' marginal costs (which determines the price in perfectly competitive markets).

(b) - Comment:

Many students only described the definitions of perfectly and imperfectly competitive market but some students did not explain very well that the strategic market players can manipulate the prices either by withholding quantity (physical withholding) or by raising the asking price (economic withholding), compared to the perfectly competitive case.

(c) - Solution:

i. In the Bertrand competition model, Company B will set its price marginally below the generation cost of Company A in order to squeeze it out from the market. The market price therefore will be:

$$\pi = 75 \left[\frac{£}{MWh}\right] \quad (25)$$

The demand level for this price will be:

$$D = \frac{250-\pi}{2} = 87.5 \left[\frac{£}{MWh}\right] \quad (26)$$

Then, all the demand will be supplied by Company B.

$$P_A = 0[MW] \quad (27)$$

$$P_B = 87.5[MW] \quad (28)$$

Finally, the profit made by companies A and B will be:

$$\Omega_A = (\pi - 75) * P_A = 0[\frac{£}{h}] \quad (29)$$

$$\Omega_B = (\pi - 62) * P_B = 1137.5[\frac{£}{h}] \quad (30)$$

ii. In the Cournot model of competition, the state of the market is determined by the production decisions made by each firm. We summarize the possible outcomes using a table where all the cells in a column corresponds to a given production by A and the cells in a row corresponds to a given production by company B.

Each cell contains four pieces of information arranged in the following format:

D	Ω_A
Ω_B	π

where:

π: price $\left[\frac{£}{MWh}\right]$

D: demand $[MW]$

Ω_A: profit made by company A $[\frac{£}{h}]$

Ω_B: profit made by company B $[\frac{£}{h}]$

Given the productions P_A and P_B of the two companies, the other quantities are calculated as follows:

$$D = P_A + P_B \quad (31)$$

$$\pi = 250 - 2 * D \quad (32)$$

$$\Omega_A = P_A * (\pi - 75) \quad (33)$$

$$\Omega_B = P_B * (\pi - 62) \quad (34)$$

Based on the above, the table expressing the Cournot model of competition for the conditions of the problem is shown below. Nash equilibrium implies that given the production level of either of the two companies in this state, the other company cannot achieve a higher profit by allowing a production level different than the one at the equilibrium. It can thus be observed form the table below that the cell corresponding to $P_A = 30MW$ and $P_B = 30MW$; $P_A = 25MW$ and $P_B = 35MW$ are two Nash equilibria.

P_B / P_A	20		25		30		35	
25	45	1700	50	1875	55	1950	60	1925
	2450	160	2200	150	1950	140	1700	130
30	50	1500	55	1625	60	1650	65	1575
	2640	150	2340	140	2040	130	1740	120
35	55	1300	60	1375	65	1350	70	1225
	2730	140	2380	130	2030	120	1680	110
40	60	1100	65	1125	70	1050	75	875
	2720	130	2320	120	1920	110	1520	100

(c) - Comment:

Many student answered part i correctly. For part ii, some students could only find one Nash equilibria point, while this question is set up for two (mixed) Nash equilibria points. The students did not summarize the possible outcomes using the table above. Furthermore, some students incorrectly calculated prices and two generators' profits.

(d) - Solution:

The cell corresponding to $P_A = 20MW$ and $P_B = 25MW$ involves higher profits than the two cells corresponding to the Nash Equilibria: $P_A = 30MW$ and $P_B = 30MW$; $P_A = 25MW$ and $P_B = 35MW$ (identified as equilibria in **(b)**). However, this cell does not satisfy the properties of NE since: a) given that $P_A = 20MW$, company B can achieve higher profits by producing $P_B = 35MW$ ($\Omega_B = 2{,}730\,£/h$) than $P_B = 25MW$ ($\Omega_B = 2{,}450\,£/h$) and also b) given that $P_B =$

$25MW$, company A can achieve higher profits by producing $P_A = 30MW$ ($\Omega_A = 1{,}950\,£/h$) than $P_A = 20MW$ ($\Omega_A = 1{,}700\,£/h$).

(d) - Comment:

Some students did not combine their analysis with the matrix table in ***(c)*** *and did not find another production level that would increase the company's profits.*

(e) - Solution:

Each generating company is trying to maximize its profit which is equal to the difference between its revenue and its cost:

$$\Omega_A = P_A * \pi(D) - C_A(P_A) \quad (35)$$

$$\Omega_B = P_B * \pi(D) - C_B(P_B) \quad (36)$$

However, we cannot treat these two maximizations as separate optimization problems since they are links through the demand:

$$D = P_A + P_B \quad (37)$$

$$\pi = 250 - 2 * D \quad (38)$$

Hence, we express the price as a function of the production of the two generators:

$$\pi = 250 - 2 * (P_A + P_B) \quad (39)$$

The profit of the two companies than be expressed as:

$$\Omega_A = P_A * (250 - 2 * P_A - 2 * P_B) - 75 * P_A = -2 * P_A^2 + (175 - 2 * P_B) * P_A \quad (40)$$

$$\Omega_B = P_B * (250 - 2 * P_A - 2 * P_B) - 62 * P_B = -2 * P_B^2 + (188 - 2 * P_A) * P_B \quad (41)$$

The optimality condition for the maximization problem of company A is:

$$\frac{\partial \Omega_A}{\partial P_A} = -4 * P_A - 2 * P_B + 175 = 0 \quad (42)$$

The optimality condition for the maximization problem of company B is:

$$\frac{\partial \Omega_B}{\partial P_B} = -4 * P_B - 2 * P_A + 188 = 0 \quad (43)$$

(We only consider the partial derivative of the profit of a generator with respect to its own output because that is the only variable over which it has control)

Putting these optimality conditions in matrix form:

$$\begin{bmatrix} -4 & -2 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} P_A \\ P_B \end{bmatrix} = \begin{bmatrix} -175 \\ -188 \end{bmatrix} \quad (44)$$

Solving this linear system we get the following productions for the NE:

$$P_A = 27[MW] \quad (45)$$

$$P_B = 33.5[MW] \quad (46)$$

Form which we can easily find:

$$\pi = 129\left[\frac{£}{MWh}\right],\ D = 60.5\ [MW],\ \Omega_A = 1{,}485[\frac{£}{h}],\ \Omega_B = 2{,}244.5[\frac{£}{h}] \quad (47)$$

(e) - Comment:

Some students treated the generation companies' profit maximizations as two separate optimization problems and did not link the two coupled optimization problems with the common demand. In summary, students did not find the relationship between the market price and dispatch of two generation companies: $\pi = 250 - 2 * (P_A + P_B)$ *through the supply demand balance constraint* $D = P_A + P_B$. *Some students did not calculate the optimal condition for profit maximization of two generating companies.*

Question 3 - Comments

Part I:

(a) - Solution:

Given that the demand we consider is 500MW and that unit 1 has the lowest marginal operating cost and cannot provide reserve, we conclude immediately that this unit must produce its maximum output of 300MW. Since units 2 and 3 are the only ones that can provide reserve and since unit 2 can provide at most 180MW, unit 3 has to provide at least 105MW. Given that unit 3 has a capacity of 270MW, its power production must be less than 165 MW.

$$0 \leq P_3 \leq 165 \quad (48)$$

Similarly, since unit 3 can provide at most 220MW of reserve, unit 2 has to provide at least 65MW. It power output is thus limited to 185 MW:

$$0 \leq P_2 \leq 185 \quad (49)$$

For the demand of 500MW, the production of unit 2 is capped at 185MW because it must provide at least 65MW of reserve. Unit 3 becomes the marginal generator and output at 15MW and provides 220MW of reserve.

(a) - Comment:

Many students answered this question correctly. No common mistakes.

(b) - Solution:

For the demand of 500MW, the price of energy is thus set at the marginal cost of unit 3, which is 21.5 £/MWh. To determine the price of reserve, we must figure out where an additional megawatt of reserve would come from and how much it would cost. Figure in **(a)** shows that if output at 15MW, unit 3 provides 220MW of reserve, which is the maximum it is able to deliver under any circumstance. To get an additional megawatt of reserve beyond the basic 285 MW requirement, we would have to reduce the output of unit 2 by one megawatt. Instead of producing 185MW, it would thus produce only 184MW. To compensate for this reduction, we would have to increase the output of unit 3 by one megawatt. This extra megawatt from unit 3 would cost £21.5,

while the reduction in the output of unit 2 would save £13. The net cost of getting an additional megawatt of reserve and hence the price of reserve is 21.5 – 13 = 8.5 £/MWh.

(b) - Comment:

Many students correctly answered the price of energy set at the marginal costs, but some did not determine the price of reserve. Students assumed the price of reserve to be the same as the price of energy at 21.5 £/MWh and did not consider the reduction in the output of unit 2 that would save £13. Thus, they did not calculate the net cost of getting an additional megawatt of reserve.

(c) - Solution:

For the demand of 500MW, even though unit 2 has a lower marginal cost than other units, its output is capped at 185MW by the reserve requirement constraint. Unit 3 is then the marginal producer. The energy price jumps from 13 to 21.5£/MWh, which means that unit 2 makes an 8.5£/MWh profit on each megawatt-hour it produces. For the demand of 500MW, unit 2 also make an 8.5£/MWh profit on every megawatt of reserve it provides. In total, the profit that unit 2 derives from both the energy and reserve markets is:

Total profit = $185 \times 8.5 + 65 \times 8.5 = 2{,}125$ £/h (50)

(c) - Comment:

Some students assumed that the output power 185MW contained the reserve requirement. Thus, the total profit calculated by these students was: 185 × 8.5 = 1,573 £/h. Furthermore, some students answered this question incorrectly because they did not calculate the reserve price correctly in ***(b).***

Part II:

(a) - Solution:

Since the engineer is only asked for a rough estimate, costs associated with maintaining the plant are not considered – also the internal rate of return (IRR) method is used to estimate the profitability of the plant. To implement this method, engineer must determine the net cash flow for each year of this plant's lifetime: the cost of building the plant:

Investment cost = $1200 \times 400 =$ £ 480,000,000 (51)

Next, the annual production of this plant should be estimated. Ideally, the plant should operate at full capacity at all times. In practice, this is not possible because the plant has to be shut down periodically for maintenance and because there will inevitably be failures resulting in unplanned outages. For the given data, we have:

Estimated annual production = $400 \times 7446 = 2{,}978{,}400$ MWh (52)

Then calculate the annual cost of producing this energy:

Annual production cost = $2{,}978{,}400 \times 9800 \times 1.1 =$ £ 32,107,152 (53)

Finally, to estimate the revenue, it can be assumed that this generating plant will be able to sell the energy it produces for 31 $/MWh. The annual revenue is thus as follows:

Annual revenue = 2,978,400 × 31 = £ 92,330,400 (54)

Annual profit = 92,330,400 − 32,107,152 = £ 60,223,248 (55)

Therefore, the total cash flow over the expected plant life (i.e. 30 years) is:

Total cash flow = 30 × 60,223,248 − 480,000,000 = £1,326,697,440 (56)

(a) - Comment:

Many students correctly calculated the investment cost, estimated annual energy production, annual production cost, annual revenue and annual profit, but did not calculate the total cash flow.

(b) - Solution:

The MARR should be set according to the risk associated with the investment. Firms that specialize in high risk investments generally set MARR high whereas firms specialize in low risk investments generally set MARR low.

Two potential issues are of particular concern for this investment:

what happens if the price of electrical energy does not meet expectation?

what if the plant cannot achieve the targeted utilization factor?

(b) - Comment:

Some students did not answer the potential risk issues correctly.

(c) - Solution:

The internal rate of return (IRR) is a metric used to estimate the profitability of potential investments. The internal rate of return is a discount rate that makes the net present value (NPV) of all cash flows from a particular project equal to zero. IRR calculations rely on the same formula as NPV does.

$$NPV = 0 = CF_0 + \frac{CF_1}{(1+IRR)} + \frac{CF_2}{(1+IRR)^2} + \cdots + \frac{CF_n}{(1+IRR)^n} = \sum_{t=0}^{n} \frac{CF_t}{(1+IRR)^t} \quad (57)$$

Borduria Power must then compare this value with their "minimum acceptable rate of return" (MARR) before they make their decision. If IRR > MARR, then the investment is OK, if IRR < MARR, then the investment is not worth making. In this case, the IRR is higher than MARR which justifies the investment in this plant.

(c) - Comment:

Some students correctly describe the process of calculation of the internal rate of return (IRR) and its relation to the net present value (NPV), but did not compare this value with their "minimum acceptable rate of return" (MARR).

Question 4 - Comments

(a) - Solution:

A brief discussion of the following points is expected:

Transmission is a natural monopoly Because of their visual impact on the environment, construction of competing transmission corridors along similar routes is not feasible. Like all monopolies that provide an essential service, electricity transmission must be regulated to ensure that it delivers an economically optimal combination of quality of service and price.

Transmission is a capital-intensive business - Making efficient investment decisions is thus critical aspect of running a transmission company.

Transmission assets have a long life - Most transmission equipment is designed for an expected life of more than 40 years. Generation and demand conditions can change over such a long period. A transmission line that is built on the basis of erroneous forecasts may therefore be used at only a fraction of its rating.

Transmission investments are irreversible Once a transmission line has been built, it cannot be redeployed cost effectively to another location - resale value of installed assets is very low. Owners of transmission networks (or regulators) therefore need to manage consequences of investment decisions for a very long time. A large investment that is not used as much as it was initially expected is called a *stranded investment*.

Transmission investments are lumpy Manufacturers sell transmission equipment in only a small number of standardized voltage levels and MVA ratings. It is therefore often not possible to build a transmission facility with rating that exactly matches the need at minimum costs.

Economies of scale Because of high fixed cost component of building transmission infrastructure, the average cost of transmitting electricity generally decreases with the amount transported.

(a) - Comment:

Most of students' answers focused on 1) Transmission is a natural monopoly; 2) Transmission assets have a long life; 3) Transmission investments are irreversible.

(b) - Solution:

A transmission system is required to efficiently and securely transport electric power from generating plants to distribution systems and large consumers so as to: a) minimize fuel costs in the production of electricity by allowing the utilization of those available generating sources having the lowest marginal costs, b) interconnect systems and generating plants to reduce overall generating requirements by taking advantage of the diversity of peak loads (peak loads occur at different times in different systems) and

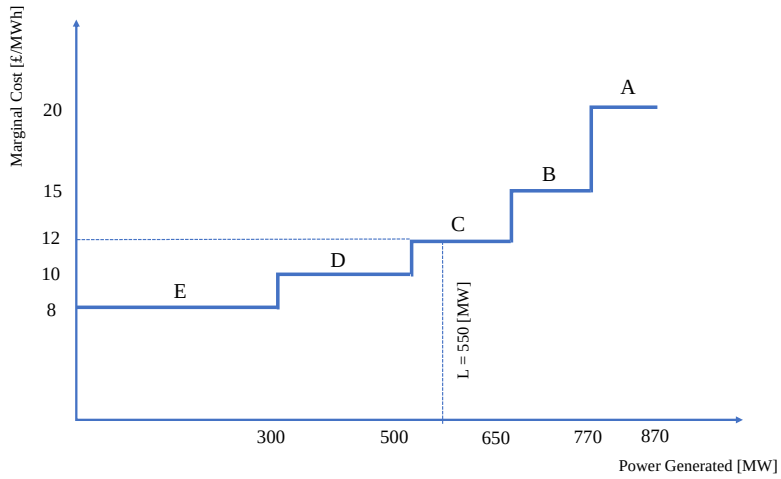
diversity of generation outages and reserve requirements and c) to enable a competitive trading of electric energy in the marketplace.

(b) - Comment:

Some students did not demonstrate the benefit of having a transmission system.

(c) - Solution:

Since there are no transmission constraints, the output of all the generators can be stacked in order of marginal cost as shown in the following figure:



Using the above figure, we can see that for a system load of $L = 400 + 100 + 50 = 550MW$ the marginal cost (and hence the nodal price) is:

$$\pi = 12\ [\frac{£}{MWh}] \quad (58)$$

Furthermore, the units are dispatched as follows:

$$P_E = 300\ [MW] \quad (59)$$

$$P_D = 200\ [MW] \quad (60)$$

$$P_C = 50\ [MW] \quad (61)$$

$$P_B = 0[MW] \quad (62)$$

$$P_A = 0[MW] \quad (63)$$

The line flow problem can be solved using the superposition principle and also directly. To this effect, we write the power balance equation at two buses and KVL around the loop.

Node 1: $P_A + P_B - 400 = F_{12} + F_{13}$ (64)

Node 2: $P_C - 100 = -F_{12} + F_{23}$ (65)

Node 3: $P_D + P_E - 50 = -F_{13} - F_{23}$ (66)

Loop equation: $0.20F_{12} - 0.25F_{13} + 0.25F_{23} = 0$ (67)

Putting these equations in the matrix form gives:

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 0.20 & -0.25 & 0.25 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} P_c - 100 \\ P_D + P_E - 50 \\ 0 \end{bmatrix} \quad (68)$$

Substituting $P_A = 0\ [MW]$, $P_B = 0[MW]$, $P_C = 50[MW]$, $P_D = 200[MW]$ and $P_E = 300[MW]$ in (69), we get:

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 0.20 & -0.25 & 0.25 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} -50 \\ 450 \\ 0 \end{bmatrix} \quad (69)$$

Solving these equations, we get:

$$F_{12} = -125\ [MW] \quad (70)$$

$$F_{13} = -275\ [MW] \quad (71)$$

$$F_{23} = -175[MW] \quad (72)$$

(c) - Comment:

Some students did not draw the marginal cost curve of each generator following the characteristic of merit order. Thus, it is not straightforward to determine the dispatch of each unit. Furthermore, some students did not give the KVL equation (67), and therefore did not calculate the matrix (68) as well as the power flows in the examined question.

(d) - Solution:

From part **c)**, we observed that the flow on line 1-3 exceeds its maximum capacity by $25[MW]$.

i) The first method consists in increasing the output of generator B and decreasing by the same amount the output of generator D. To calculate how big this increase should be to remove the violation of the flow limit on line 3-1, consider an injection of $+1[MW]$ at bus 1 and an injection of $-1[MW]$ at bus 3. This pair of injection causes a flow in the network that divides itself as follows:

$$\frac{0.25}{(0.20+0.25)+0.25} \cdot 1 = 0.3571[MW] \text{ along the path 1-2-3} \quad (73)$$

$$\frac{(0.20+0.25)}{(0.20+0.25)+0.25} \cdot 1 = 0.6429[MW] \text{ along the path 1-3} \quad (74)$$

Since we use a linear (dc) model, we can say that to remove $25[MW]$ overload on line 3-1, we therefore need to increase the output of generator B by:

$$\frac{25}{0.6429} = 38.89\ [MW] \quad (75)$$

The constrained dispatch is then:

$$P_A = 0\ [MW] \quad (76)$$

$$P_B = 0 + 38.89 = 38.89\ [MW] \quad (77)$$

$$P_C = 50[MW] \quad (78)$$

$$P_D = 200 - 38.89 = 161.11[MW] \quad (79)$$

$$P_E = 300[MW] \quad (80)$$

Using the nodal and loop equations, the flows are calculated solving the following linear system:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & -1 \\ 0.20 & -0.25 & 0.25 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} P_A + P_B - 400 \\ P_D + P_E - 50 \\ 0 \end{bmatrix} \quad (81)$$

Substituting $P_A = 0\,[MW]$, $P_B = 38.89[MW]$, $P_C = 50[MW]$, $P_D = 161.11[MW]$and $P_E = 300[MW]$ in (81), we get:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & -1 \\ 0.20 & -0.25 & 0.25 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} -361.11 \\ 411.11 \\ 0 \end{bmatrix} \quad (82)$$

we get:

$F_{12} = -111.11\,[MW]$ (83)

$F_{13} = -250[MW]$ (84)

$F_{23} = -161.11\,[MW]$ (85)

Then, we can see that this dispatch does not cause a violation of the line flow constrains on any other line.

The cost of this dispatch is:

$C_{Total} = 38.89 * 15 + 50 * 12 + 161.11 * 10 + 300 * 8 = 5194.45\,[£]$ (86)

ii) The other method to remove the constraint violation consists in increasing the output of generator C and decreasing the output of generator D by the same amount. To calculate how big this increase should be to remove the violation of the flow on line 3-1, consider an injection of $+1[MW]$ at bus 2 and an injection of $-1[MW]$ at bus 3. This pair of injection causes a flow in the network that divides itself as follows:

$\frac{0.25}{(0.20+0.25)+0.25} \cdot 1 = 0.3571\,[MW]$ along the path 2-1-3 (87)

$\frac{(0.20+0.25)}{(0.20+0.25)+0.25} \cdot 1 = 0.6429\,[MW]$ along the path 2-3 (88)

Since we use a linear (dc) model, we can say that to remove $25[MW]$ overload on line 3-1, we therefore need to increase the output of generator C by:

$\frac{25}{0.3571} = 70\,[MW]$ (89)

The constrained dispatch is then:

$P_A = 0\,[MW]$ (90)

$P_B = 0\,[MW]$ (91)

$P_C = 50 + 70 = 120[MW]$ (92)

$P_D = 200 - 70 = 130[MW]$ (93)

$P_E = 300[MW]$ (94)

Using the nodal and loop equations, the flows are calculated solving the following linear system:

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 0.20 & -0.25 & 0.25 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} P_C - 100 \\ P_D + P_E - 50 \\ 0 \end{bmatrix} \quad (95)$$

Substituting $P_A = 0\ [MW]$, $P_B = 0[MW]$, $P_C = 120[MW]$,$P_D = 130[MW]$and $P_E = 300[MW]$ in (95), we get:

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 0.20 & -0.25 & 0.25 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} 20 \\ 380 \\ 0 \end{bmatrix} \quad (96)$$

we get:

$$F_{12} = -150\ [MW] \quad (97)$$

$$F_{13} = -250\ [MW] \quad (98)$$

$$F_{23} = -130[MW] \quad (99)$$

Once again, this redispatch does not cause a violation of the line flow constrains on any other line. The cost of this constrained dispatch is:

$$C_{Total} = 120 \cdot 12 + 130 \cdot 10 + 300 \cdot 8 = 5140\ [£] \quad (100)$$

Even though it re-dispatches a larger amount of MW, the second constrained dispatch is preferable to the first one because its cost is lower. It is thus the optimal constrained dispatch.

(d) - Comment:

The biggest issue was that many students did not calculate how big this increase should be to remove the violation of the flow on line 3-1. In other words, these students did not apply the principle of KVL in the mesh network. Furthermore, some students providing the correct equation but incorrectly calculated the power flow matrix. Most of students calculated correctly the cost of the constrained dispatch.

(e) - Solution:

The nodal price at each bus is given by the cost of one additional MW of load at each node. Therefore, the price at bus 3 is $10\left[\frac{£}{MWh}\right]$ because the next MW of load would be generated locally by generator D because it is the cheapest generator not operating at its upper limit. An additional MW of load at node 2 would have to be produced by generator C. Producing it with generator D would cause a violation of the line flow constraint on line 3-1. Producing it with generators A and B would be more expensive than with generator C. The price at node 2 is therefore $12\left[\frac{£}{MWh}\right]$. An additional MW of load at bus 1 requires a re-dispatch of generators C and D to minimize the cost increase while maintaining the flow on line 3-1 within limits.

Extracting an additional $1[MW]$ at bus 1 and generating it at bus 3 causes the following change in the flow on line 1-3:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & -1 \\ 0.20 & -0.25 & 0.25 \end{bmatrix} \begin{bmatrix} \Delta F_{12} \\ \Delta F_{13} \\ \Delta F_{23} \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \Longrightarrow \Delta F_{31} = 0.6429\ [MW] \quad (101)$$

Similarly, extracting an additional $1[MW]$ at bus 1 and generating it at bus 2 causes the following change in the flow on line 1-3:

$$\begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0.20 & -0.25 & 0.25 \end{bmatrix} \begin{bmatrix} \Delta F_{12} \\ \Delta F_{13} \\ \Delta F_{23} \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \Longrightarrow \Delta F_{31} = 0.2857\ [MW] \quad (102)$$

Therefore, if we want that the flow on line 3-1 remains unchanged (because it is already at its limit), we must change the productions by generators C and D in such a way that:

$$\Delta F_{31} = 0 = 0.6429 \cdot \Delta P_D + 0.2857 \cdot \Delta P_C \quad (103)$$

At the same time, since we are increasing the load by $1[MW]$, we must also have:

$$\Delta P_C + \Delta P_D = 1 \quad (104)$$

Solving the system consisting of the previous two equations we get:

$$\Delta P_D = -0.8\ [MW] \quad (105)$$

$$\Delta P_C = 1.8\ [MW] \quad (106)$$

To supply an additional MW of load at bus 1 without violating the network constraints, we must therefore increase the output of generator C and decrease the output of generator D.

The nodal price at bus 1 is thus given by a linear combination of the marginal cost of production of these two generators.

$$\pi_1 = -0.8 \cdot 10 + 1.8 \cdot 12 = 13.6\ [\tfrac{£}{MWh}] \quad (107)$$

Finally:

$$\pi_1 = 13.6[\tfrac{£}{MWh}] \quad (108)$$

$$\pi_2 = 12[\tfrac{£}{MWh}] \quad (109)$$

$$\pi_3 = 10[\tfrac{£}{MWh}] \quad (110)$$

(e) - Comment:

*Most of the students that successfully answered question **(d)**, also correctly answered this question. Students that did not apply the concept of KVL, did not calculate the power flow matrix (101)-(102), and consequently the nodal prices.*